

Notes: Baqaee & Farhi (QJE, 2020)

Productivity and Misallocation in General Equilibrium

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1 Big picture

- **Question.** How should we aggregate microeconomic productivity shocks in an economy with distortions? Hulten’s theorem says that in an efficient economy aggregate TFP changes are sales-share-weighted averages of micro TFP changes. This paper asks what replaces that result once the economy has markups, taxes, financial frictions, or other wedges.
- **Core contribution.** The paper derives a general first-order aggregation formula for inefficient economies with arbitrary neoclassical production, input-output networks, multiple factors, and arbitrary wedges. The key conceptual move is to rely on *micro envelope conditions* (cost minimization at the producer level) instead of the *macro envelope condition* that comes from the first welfare theorem.
- **Main decomposition.** The impact of a micro shock on aggregate output or aggregate TFP can be split into:
 - a **pure technology effect**: what happens if technology changes holding the allocation of resources fixed; and
 - a **change in allocative efficiency**: what happens because resources are reallocated toward or away from more distorted parts of the economy.
- **Why this matters.** In efficient economies, reallocations do not matter to first order. In inefficient economies, they do. So the aggregate effect of a micro shock depends not just on who becomes more productive, but also on whether the shock reallocates inputs toward high-markup or low-markup supply chains.
- **Measurement contribution.** The paper shows that in distorted economies the usual Solow residual is not the right measure of aggregate TFP growth. The correct object is a **distortion-adjusted Solow residual** that weights factor growth by *cost-based* factor shares rather than observed revenue shares.
- **Counterfactual contribution.** The paper also derives a general formula for the **distance to the frontier**—the TFP gain from eliminating wedges. This extends the Hsieh–Klenow logic to arbitrary input-output networks and general substitution patterns.
- **Empirical application.** Using U.S. firm-level markup data, the paper finds that from 1997 to 2015 improvements in allocative efficiency due to reallocation account for about **half** of aggregate TFP growth. It also finds that eliminating markups would raise aggregate TFP by roughly **10%–25%**, with a benchmark estimate around **15%**.
- **Deep intuition.** If high-markup firms are socially “too small” relative to the efficient allocation, then reallocation toward those firms can *raise measured TFP growth over time*, even though a higher dispersion of markups can simultaneously increase the economy’s *distance from the efficient frontier*. This is the paper’s most important conceptual subtlety.

2 Data and measurement (theoretical objects and empirical inputs)

2.1 Theoretical environment and key objects

- **Producers and factors.** There are N producers and F factors with inelastic supplies L_f . Producers use intermediate inputs and factors, sell to one another, and sell to final demand.

- **Wedges.** The primitive environment allows taxes and wedges on goods and factors, but the paper shows that these can be represented as markups charged by fictitious producers. So for most of the analysis all distortions are summarized by a vector of wedges/markups μ .
- **Variable returns and non-Hicks-neutral shocks.** These can also be nested into the framework. Variable returns can be represented by fixed factors; non-Hicks-neutral shocks can be represented by introducing fictitious producers.
- **Output concept.** Real GDP is defined using a constant-returns final-demand aggregator. That lets the paper talk cleanly about output and aggregate TFP without index-number ambiguities.

2.2 Input-output objects that drive the theory

For readability, I will write factor revenue shares as s_f and cost-based factor shares as $\tilde{s}_f$. These correspond to the paper's factor Domar-weight notation.

- **Final expenditure shares:**

$$b_i = \frac{p_i c_i}{\sum_{j=1}^N p_j c_j}.$$

- **Revenue-based input-output matrix:**

$$\Gamma_{ij} = \frac{p_j x_{ij}}{p_i y_i}.$$

This is the expenditure share of producer i on input j measured relative to i 's revenues.

- **Cost-based input-output matrix:**

$$\tilde{\Gamma}_{ij} = \frac{\partial \log C_i}{\partial \log p_j} = \frac{p_j x_{ij}}{\sum_k p_k x_{ik}}.$$

This is the share of input j in producer i 's total costs.

- **Link between the two:**

$$\tilde{\Gamma} = \text{diag}(\mu) \Gamma.$$

When $\mu_i > 1$, the producer's cost shares exceed its revenue expenditure shares because price exceeds marginal cost.

- **Leontief inverses:**

$$\Psi = (I - \Gamma)^{-1}, \quad \tilde{\Psi} = (I - \tilde{\Gamma})^{-1}.$$

Ψ propagates *demand* backward through the network; $\tilde{\Psi}$ propagates *costs* forward.

- **Revenue-based Domar weights:**

$$\lambda_i = \frac{p_i y_i}{\sum_{j=1}^N p_j c_j}, \quad \lambda' = b' \Psi.$$

- **Cost-based Domar weights:**

$$\tilde{\lambda}' = b' \tilde{\Psi}.$$

These are the right weights for pure technology effects in distorted economies.

2.3 Empirical application: U.S. markups and production networks

- **Input-output data.** The paper uses annual U.S. BEA input-output tables for 1997–2015 with 66 industries, dropping government, noncomparable imports, and second-hand scrap.
- **Firm-level data.** The firm side comes from Compustat. Firms are assigned to BEA industries and assumed to produce a single output.
- **Within-industry structure.** The outputs of firms within an industry are aggregated homothetically into an industry bundle. Firms in the same industry are assumed to share the same production function up to Hicks-neutral productivity shifts.
- **Key imputation.** The firm-level network is not observed. The paper reconstructs it from industry-level input-output data plus the within-industry distribution of firm sales and markups.

The mapping from industry-level data to firm-level objects is:

$$\mu_I = \left(\sum_{i \in I} \frac{\lambda_i}{\lambda_I} \mu_i^{-1} \right)^{-1}, \quad (1)$$

$$\tilde{\Gamma}_{IJ} = \mu_I \Gamma_{IJ}, \quad \tilde{\Gamma}_{ij} = \frac{\lambda_j}{\lambda_J} \tilde{\Gamma}_{IJ} \quad \text{for } i \in I, j \in J, \quad (2)$$

$$\Gamma_{ij} = \mu_i^{-1} \tilde{\Gamma}_{ij}. \quad (3)$$

- **Markup series.** The paper uses three firm-level markup measures:
 - **UC (user-cost):** benchmark series; average markup around 5%;
 - **AP (accounting profits):** average markup around 10%;
 - **PF (production function):** average markup around 16%.
- **Empirical pattern that matters most.** Across all three series, the rise in average markup is driven mostly by a **between-firm composition effect**: high-markup firms get larger, rather than individual firms sharply increasing their own markups.
- **Why that matters for the theory.** Reallocation toward high-markup firms can raise measured TFP growth if those firms were too small relative to the efficient allocation, even though greater markup dispersion raises the static distance to the frontier.

3 Models and empirical specifications (all equations + interpretation)

3.1 Final demand and the producer problem

Final demand is defined by the constant-returns aggregator

$$Y = \max_{\{c_1, \dots, c_N\}} D(c_1, \dots, c_N) \quad (4)$$

subject to

$$\sum_{i=1}^N (1 + \tau_i^0) p_i c_i = \sum_{f=1}^F w_f L_f + \sum_{i=1}^N \pi_i + \tau. \quad (5)$$

Producer i has cost function

$$\frac{1}{A_i} C_i \left((1 + \tau_{i1}) p_1, \dots, (1 + \tau_{iN}) p_N, (1 + \tau_{i1}^f) w_1, \dots, (1 + \tau_{iF}^f) w_F \right) y_i \quad (6)$$

and charges price

$$p_i = \mu_i \frac{C_i}{A_i}. \quad (7)$$

- **Interpretation.** A_i is Hicks-neutral productivity; μ_i summarizes wedges/markups; C_i is the unit-cost object dual to the production technology.
- **Economic point.** The paper allows arbitrary neoclassical production and arbitrary wedges, but it keeps one crucial structure: each producer cost-minimizes given prices. That micro envelope condition is what ultimately powers the aggregation formulas.

3.2 Main ex post aggregation theorem

The central first-order result is

$$\frac{\partial \log Y}{\partial \log A_k} = \tilde{\lambda}_k - \sum_f \tilde{s}_f \frac{\partial \log s_f}{\partial \log A_k}, \quad (8)$$

$$\frac{\partial \log Y}{\partial \log \mu_k} = -\tilde{\lambda}_k - \sum_f \tilde{s}_f \frac{\partial \log s_f}{\partial \log \mu_k}, \quad (9)$$

and therefore

$$\Delta \log Y = \tilde{\lambda}' \Delta \log A - \tilde{\lambda}' \Delta \log \mu - \tilde{s}' \Delta \log s. \quad (10)$$

- **Pure technology effect.** $\tilde{\lambda}' \Delta \log A$. This is the effect of technology holding the allocation matrix fixed. The weights are *cost-based* Domar weights, not revenue-based ones.
- **Allocative-efficiency effect.** $-\tilde{\lambda}' \Delta \log \mu - \tilde{s}' \Delta \log s$. This is the effect of reallocation: the output change generated because shocks move resources across distorted uses.
- **Main takeaway.** In a distorted economy the aggregate effect of a micro shock is not just “Domar weight times micro shock.” You must add how the shock changes factor shares and hence the allocation of resources.

3.3 Hulten as a special case

If the initial equilibrium is efficient ($\mu = 1$), then

$$\frac{\partial \log Y}{\partial \log A_k} = \lambda_k, \quad \frac{\partial \log Y}{\partial \log \mu_k} = 0. \quad (11)$$

- **Interpretation.** Hulten’s theorem is a special case in which the allocative-efficiency term vanishes to first order because the initial allocation is already efficient.

- **Why the generalization is economically meaningful.** The paper is not merely tweaking a formula; it is showing exactly where the first welfare theorem enters Hulten’s result and what replaces it when that theorem fails.

3.4 Illustrative special cases: vertical, horizontal, and round-about economies

The three diagrams in Figure I are there to separate two different ways Hulten can fail:

- revenue-based and cost-based Domar weights can differ; and
- reallocations can matter to first order.

For the **horizontal economy**, the paper obtains

$$\frac{d \log Y}{d \log A_k} = \lambda_k - \lambda_k(\theta_0 - 1) \left(\frac{\mu_k^{-1}}{\sum_j \lambda_j \mu_j^{-1}} - 1 \right), \quad (12)$$

$$\frac{d \log Y}{d \log \mu_k} = \lambda_k \theta_0 \left(\frac{\mu_k^{-1}}{\sum_i \lambda_i \mu_i^{-1}} - 1 \right). \quad (13)$$

- **Interpretation of the productivity formula.** A productivity shock does the usual pure technology job, but it also changes relative prices and therefore reallocates labor across producers. Whether that improves allocative efficiency depends on the substitution elasticity θ_0 and whether producer k ’s markup is above or below the harmonic-average markup.
- **Interpretation of the markup formula.** A markup increase acts like a negative productivity shock in demand space: it pushes resources away from producer k . Whether that helps or hurts output depends on whether k was socially too small or too large.
- **Benchmark elasticities.** For productivity shocks, Cobb-Douglas ($\theta_0 = 1$) kills reallocations. For markup shocks, Leontief ($\theta_0 = 0$) kills reallocations.

3.5 Distortion-adjusted Solow residual

Once factor supplies are allowed to move, the right TFP residual is

$$\Delta \log Y_t - \tilde{s}'_{t-1} \Delta \log L_t. \quad (14)$$

The paper’s decomposition is

$$\Delta \log Y_t - \tilde{s}'_{t-1} \Delta \log L_t \approx \tilde{\lambda}'_{t-1} \Delta \log A_t - \tilde{\lambda}'_{t-1} \Delta \log \mu_t - \tilde{s}'_{t-1} \Delta \log s_t. \quad (15)$$

- **Interpretation.** Aggregate TFP growth is output growth net of the *pure technology* effect of factor growth, where the factor-growth weights are cost-based, not revenue-based.
- **Why the usual Solow residual fails.** In distorted economies factor income shares can move for reasons unrelated to true aggregate technology. The traditional residual incorrectly reads some wedge-driven redistribution as TFP.
- **Empirical usefulness.** This gives a workable first-order decomposition of aggregate TFP growth into technology versus allocative efficiency using observable factor-share changes plus estimated markups.

3.6 Alternative decompositions and the acyclic-economy critique

The paper argues that alternative decompositions are conceptually flawed because they are not built from feasible counterfactual allocations.

Their pure-technology terms are:

$$\sum_k \lambda_{kt} \Delta \log A_{kt} \quad (\text{Petrin-Levinsohn}), \quad (16)$$

and

$$\sum_k \lambda_{kt} \frac{1 - s_{kt}^M}{1 - \mu_{kt} s_{kt}^M} \Delta \log A_{kt} \quad (\text{Basu-Fernald}). \quad (17)$$

The paper’s critique becomes sharp in an **acyclic economy**, where there is only one feasible allocation. In that case,

$$\frac{d \log Y}{d \log A_k} = \tilde{\lambda}_k, \quad \frac{d \log Y}{d \log \mu_k} = 0. \quad (18)$$

- **Interpretation.** If there is only one feasible allocation, there is literally no room for reallocations. So any decomposition that still reports reallocative effects in that environment is conceptually wrong.
- **Why their decomposition is better.** The paper’s pure-technology term corresponds to a feasible counterfactual allocation holding the allocation matrix fixed. Competing decompositions group residuals algebraically, but do not map to feasible allocations.

3.7 Ex ante structural results in nested CES economies

The paper then moves from *ex post* reduced-form sufficient statistics to *ex ante* structural expressions that link the factor-share response to elasticities of substitution and network structure.

In the nested CES “standard form,” each CES nest is treated as a separate producer. Let θ_j be the elasticity attached to row j . Define the input-output covariance operator

$$\text{Cov}_j(a, b) = \sum_i \tilde{\Gamma}_{ji} a_i b_i - \left(\sum_i \tilde{\Gamma}_{ji} a_i \right) \left(\sum_i \tilde{\Gamma}_{ji} b_i \right). \quad (19)$$

Then the factor-share response to a productivity shock satisfies

$$\frac{\partial \log s_f}{\partial \log A_k} = \sum_j \frac{\lambda_j}{\mu_j} (\theta_j - 1) \text{Cov}_j \left(\tilde{\Psi}^{(k)} - \sum_g \tilde{\Psi}^{(g)} \frac{\partial \log s_g}{\partial \log A_k}, \frac{\Psi^{(f)}}{s_f} \right). \quad (20)$$

The paper rewrites this as the linear system

$$\frac{\partial \log s}{\partial \log A_k} = \Phi \frac{\partial \log s}{\partial \log A_k} + \delta^{(k)}, \quad (21)$$

where $\delta^{(k)}$ is the **factor-share impulse** from shock k and Φ is the **factor-share propagation matrix**.

For markup shocks,

$$\frac{\partial \log s_f}{\partial \log \mu_k} = - \sum_j \frac{\lambda_j}{\mu_j} (\theta_j - 1) \text{Cov}_j \left(\tilde{\Psi}^{(k)} + \sum_g \tilde{\Psi}^{(g)} \frac{\partial \log s_g}{\partial \log \mu_k}, \frac{\Psi^{(f)}}{s_f} \right) - \lambda_k \frac{\Psi_{kf}}{s_f}. \quad (22)$$

- **Interpretation.** These formulas say that reallocations depend on three ingredients:
 - how a shock changes relative prices through the network ($\tilde{\Psi}$);
 - how substitution elasticities translate relative-price changes into expenditure shifts (θ_j);
 - how distorted the destination supply chains are ($\Psi^{(f)}/s_f$ -type terms).
- **Single-factor intuition.** With one factor, the expressions simplify sharply. A productivity shock reduces the labor share when it induces substitution toward more distorted supply chains. A markup shock reduces the labor share both mechanically and through reallocation; to isolate allocative efficiency, the mechanical part must be netted out.

3.8 Benchmark cases: Cobb-Douglas and Leontief

The paper emphasizes two benchmark cases:

$$\theta_j = 1 \ \forall j \implies \frac{d \log Y}{d \log A_k} = \tilde{\lambda}_k \quad \text{and no reallocative effect from productivity shocks,} \quad (23)$$

and

$$\theta_j = 0 \ \forall j \implies \frac{d \log Y}{d \log \mu_k} = 0 \quad \text{and no reallocative effect from markup shocks.} \quad (24)$$

- **Cobb-Douglas logic.** If all elasticities are unitary, productivity shocks do not change expenditure shares. So the allocation matrix is invariant to A .
- **Leontief logic.** If all elasticities are zero, wedges do not change physical allocation because input proportions are fixed. So markup shocks do not move output.

3.9 Distance to the frontier

The efficient frontier is $Y^* = Y(A, 1)$. The distance to the frontier is

$$\mathcal{L} = \log \left(\frac{Y^*}{Y} \right). \quad (25)$$

For small wedges $\mu_k = \exp(\Delta \log \mu_k)$, the second-order approximation is

$$\begin{aligned} \mathcal{L} \approx & \frac{1}{2} \sum_j \lambda_j \theta_j \text{Var}_{\Gamma(j)} \left(\sum_k \Psi^{(k)} \Delta \log \mu_k \right) \\ & + \frac{1}{2} \sum_j \lambda_j \theta_j \text{Cov}_{\Gamma(j)} \left(\sum_g \Psi^{(g)} \Delta \log s_g, \sum_l \Psi^{(l)} \Delta \log \mu_l \right). \end{aligned} \quad (26)$$

With a single factor, this simplifies to

$$\mathcal{L} \approx \frac{1}{2} \sum_j \lambda_j \theta_j \text{Var}_{\Gamma(j)} \left(\sum_k \Psi^{(k)} \Delta \log \mu_k \right). \quad (27)$$

For the horizontal economy, the familiar Hsieh–Klenow style formula becomes

$$\mathcal{L} \approx \frac{1}{2} \theta_0 \text{Var}_\lambda (\Delta \log \mu). \quad (28)$$

- **Interpretation.** Misallocation rises with markup dispersion, rises with substitution elasticities, and is amplified by the network because wedges propagate through intermediate-input linkages.
- **Important conceptual distinction.** A rise in measured allocative efficiency *over time* need not mean the economy is getting closer to the frontier. Reallocation can raise actual output relative to last period even while the efficient frontier moves further away because markup dispersion increases.

3.10 Endogenous productivities and wedges

If deeper primitives Z move both productivity and wedges, the paper applies the chain rule:

$$\frac{d \log Y}{dZ} = \sum_k \frac{d \log Y}{d \log A_k} \frac{d \log A_k}{dZ} + \sum_k \frac{d \log Y}{d \log \mu_k} \frac{d \log \mu_k}{dZ}. \quad (29)$$

- **Interpretation.** Their formulas remain useful even when markups are endogenous, as long as you can separately characterize how Z moves A and μ .
- **Use case.** This is how the framework can be embedded inside richer models of oligopoly, innovation, monetary nonneutrality, or endogenous wedge determination.

4 Identification, implementation, and what makes the application credible

4.1 What makes the theory credible

- **Key discipline.** The paper is not identified by an instrument or a natural experiment. It is identified by equilibrium accounting plus producer cost minimization.
- **Micro vs macro envelopes.** Hulten uses the macro envelope that comes from efficiency. Baqaee–Farhi replace it with micro envelopes that survive even in distorted economies. That is the central theoretical source of credibility.
- **Strength of the result.** Once the model primitives are given, the decomposition is exact to first order. The main empirical uncertainty is therefore not “is the algebra right?” but “are the wedges and network measured correctly?”

4.2 Implementation difficulties emphasized by the authors

- **Observability.** b , Γ , and revenue-based Domar weights are directly observable. By contrast, $\tilde{\Gamma}$, $\tilde{\Psi}$, and $\tilde{\lambda}$ require knowledge of wedges:

$$\tilde{\Gamma} = \text{diag}(\mu)\Gamma, \quad \tilde{\Psi} = (I - \tilde{\Gamma})^{-1}, \quad \tilde{\lambda}' = b'\tilde{\Psi}.$$

- **Factor identification.** In distorted economies one must decide what counts as a true factor payment and what is profit. Quasi-fixed factors are especially hard because they create decreasing returns and blur the line between rents and factor income.

- **Aggregation level.** This is one of the paper’s biggest implementation points. In efficient economies aggregate shocks can often be handled with aggregated sales data. In inefficient economies cost-based Domar weights depend on wedges at the level where wedges appear. If wedges are at the firm level, sector-level data can be badly misleading.
- **Mapping wedges to data frictions.** The paper spends real effort showing how nonpecuniary or non-tax frictions must be converted into the accounting convention used by the model. Financial frictions and compensating differentials can be represented with fictitious producers, but the empirical shares must be adjusted carefully.

4.3 Why the empirical application is suggestive but not final

- **Firm-level network is imputed.** The model ideally wants the network observed at the same level as the wedge. In practice the authors impute a firm-level network from industry-level data.
- **Compustat coverage.** The firm sample is publicly listed firms, which may not represent the whole economy.
- **Only markups are modeled as wedges.** Other frictions could coexist and interact with markups. The application assumes they do not dominate the aggregate picture.
- **But the qualitative message is robust.** The paper gets the same broad story from three very different markup series and shows that the central result disappears when the data are aggregated to the sector level. That pattern is exactly what the theory predicts.

5 Tables and figures

5.1 Figure I: Vertical, horizontal, and round-about economies

Read:

- This is the paper’s intuition figure. It shows three toy economies designed to isolate the two ways Hulten’s theorem can fail.
- **Vertical economy:** the allocation is still efficient because there is only one feasible allocation, but revenue-based and cost-based Domar weights differ because of downstream double marginalization.
- **Horizontal economy:** revenue-based and cost-based Domar weights coincide, but reallocations matter because a productivity or markup shock moves labor across producers with different wedges.
- **Round-about economy:** both failures occur at once. This is the closest miniature version of the general distorted network problem.

5.2 Figure II: Acyclic economies

Read:

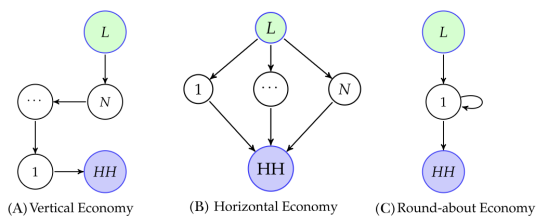


FIGURE I

Vertical, Horizontal, and Round-About Economies

The green and blue nodes (L and HH respectively; color version available online) represent the factor and the household. The arrows represent the flow of goods.

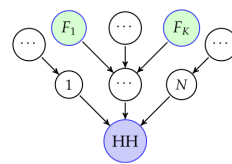


FIGURE II

Acyclic Economies

The green and blue nodes (F and HH respectively; color version available online) represent the factors and the household. The arrows represent the flow of goods.

- The purpose of this figure is not realism but logic. In an acyclic network every factor and good has a unique user, so the feasible allocation is unique.
- That means there cannot be first-order reallocative effects. If a decomposition still finds them here, it is conceptually suspect.
- This figure is the paper's cleanest critique of Basu–Fernald and Petrin–Levinsohn: those decompositions can mechanically manufacture reallocative effects even when no reallocation is feasible.

5.3 Figure III: Decomposition of the growth in the average markup

Read:

- The average markup rises over time, but the important result is *why*: the increase is driven mostly by the **between-firm** component, not the within-firm component.
- That means high-markup firms are growing in market share. This is the applied fact that later maps directly into a positive allocative-efficiency contribution.
- The figure is therefore the bridge between the abstract theory and the U.S. application: measured markups rise because the composition of firms changes.

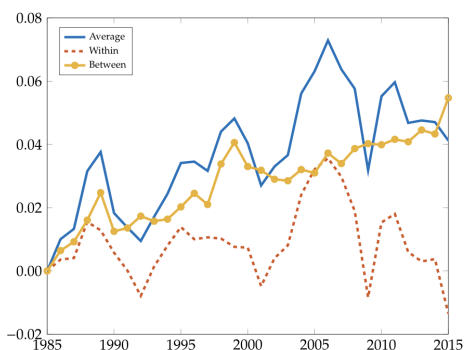


FIGURE III

Decomposition of the Growth in the Average Markup

Decomposition of the increase in the average markup into a between and within effect, using the user-cost approach markup data. All the changes are cumulated over time. See footnote 38 for more information.

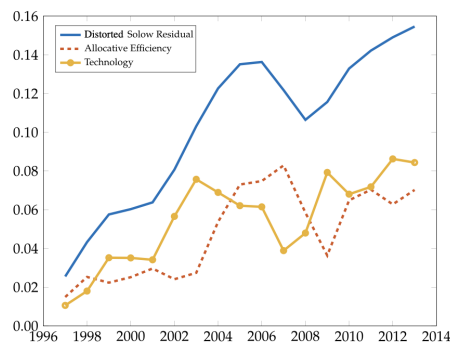


FIGURE IV

Decomposition of Aggregate TFP Growth (Distortion-Adjusted Solow Residual)

Decomposition of changes in aggregate TFP into pure changes in technology and changes in allocative efficiency following equation (7), with markups obtained using the user-cost approach. All the changes are cumulated over time.

5.4 Figure IV: Decomposition of aggregate TFP growth

Read:

- This is the paper's headline empirical figure. Using the benchmark UC markups, the cumulative distortion-adjusted Solow residual rises steadily from 1997 to 2015.
- Roughly half of that cumulative rise is attributed to **changes in allocative efficiency**, with the other half attributed to **pure technology**.
- The figure operationalizes equation (7). It is the main quantitative claim that reallocations toward high-markup firms contributed materially to measured aggregate TFP growth.

5.5 Figure V: Decomposition of aggregate TFP growth with sectoral data

Read:

- This figure repeats Figure IV after first aggregating firms to the sector level.
- The allocative-efficiency contribution becomes tiny and even slightly negative. So the strong result in Figure IV is a **within-industry firm-level phenomenon**.
- This is exactly the implementation warning from Section VI.C: in distorted economies aggregation is not innocuous.

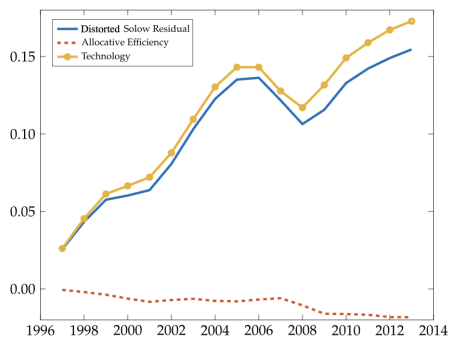


FIGURE V

Decomposition of Aggregate TFP Growth (Distortion-Adjusted Solow Residual) with Sectoral Data

This decomposition is the same as the one in Figure IV, but the markup data has first been aggregated at the industry level. Intuitively, this assumes that each industry contains a representative firm (ruling out reallocations within industries).

TABLE I GAINS FROM ELIMINATING MARKUPS FOR THE BEGINNING AND END OF OUR SAMPLE			
	User cost (UC)	Accounting (AP)	Production function (PF)
2015	13%	11%	25%
1997	3%	5%	23%

5.6 Table I: Gains from eliminating markups in 1997 and 2015

Read:

- This table reports the distance to the frontier for the markup application.
- In 2015, eliminating markups raises aggregate TFP by about 13% using UC markups, 11% using AP markups, and 25% using PF markups.
- In 1997, the corresponding gains are much smaller for UC and AP (3% and 5%), but still large for PF (23%).

TABLE II
GAINS IN AGGREGATE TFP FROM ELIMINATING MARKUPS FOR DIFFERENT MARKUP
SERIES AND DIFFERENT CALIBRATIONS

	Benchmark	CD + CES	$\xi = 4$	Cobb-Douglas	No IO	Sectoral
UC	13%	14%	8%	3%	5%	0.7%
AP	11%	12%	6%	3%	5%	1%
PF	25%	29%	14%	10%	14%	4%

Notes. The first column are the benchmark numbers in Table I. The second column sets all elasticities except ξ equal to 1. The third column lowers the elasticity of substitution ξ between firms within each industry. The fourth column sets all elasticities in the model to be equal to 1. The fifth column recalibrates the model using value-added production functions, ignoring the input-output connections, similar to [Haieh and Klenow \(2009\)](#). The final column uses data aggregated to the sectoral level rather than firm level, similar to [Harberger \(1954\)](#).

- The table’s punchline is twofold: the static losses from markup dispersion are large, and they generally became larger over time.

5.7 Table II: Robustness to alternative structural calibrations

Read:

- This table shows that the measured deadweight loss is very sensitive to the details that Harberger-style calculations abstract from.
- Lowering the within-industry elasticity ξ from 8 to 4 roughly halves the estimated loss. That matches the theory: with less substitution across firms, wedges cause less reallocation.
- Moving to a fully Cobb-Douglas model cuts the losses sharply. Ignoring input-output linkages also cuts them sharply. Aggregating to the sector level cuts them dramatically, down to numbers much closer to classic Harberger calculations.
- The economic lesson is that **micro-level dispersion**, **realistic substitution elasticities**, and **production networks** are all quantitatively first-order for markup misallocation.

5.8 Figures A1 and A2: Robustness across PF and AP markup series

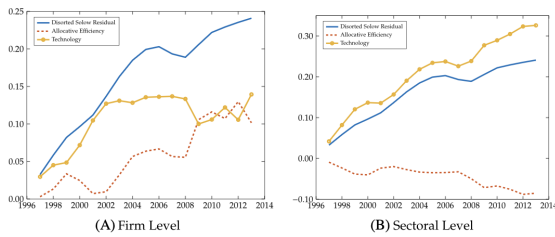


FIGURE A.1

Decomposition of Aggregate TFP Growth (Distortion-Adjusted Solow Residual)
with PF Markups

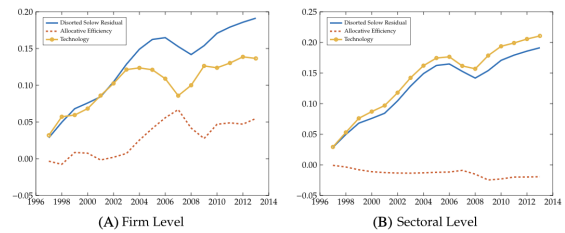


FIGURE A.2

Decomposition of Aggregate TFP Growth (Distortion-Adjusted Solow Residual)
with AP Markups

Read:

- These appendix figures repeat the TFP decomposition using the PF and AP markup measures.
- The exact magnitudes differ, but the qualitative pattern remains: at the firm level allocative efficiency contributes positively and materially to cumulative TFP growth, while the effect weakens sharply after sectoral aggregation.

- This is important because the three markup measures are conceptually quite different. The robustness suggests the key pattern is not an artifact of one particular markup estimator.

6 Short summary

- **Conclusion.** In inefficient economies aggregate TFP growth must be decomposed into a pure technology term and a first-order allocative-efficiency term; the standard Hulten/Solow formulas are not enough.
- **Intuition.** A shock changes output directly through production, but it also changes where factors and intermediate inputs go. In a distorted economy that reallocation itself changes productivity.
- **Empirical lesson.** Using U.S. firm-level markups, the paper argues that reallocations toward high-markup firms account for about half of measured aggregate TFP growth since the late 1990s, even though markup dispersion simultaneously increases the distance from the efficient frontier.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main theoretical result.** The aggregate effect of micro shocks in a distorted economy is

$$\text{technology} + \text{reallocation},$$

where the technology term is weighted by cost-based Domar weights and the reallocation term is summarized by changes in factor shares.

- **Measurement result.** The right aggregate TFP measure is the distortion-adjusted Solow residual, not the standard Solow residual.
- **Counterfactual result.** The paper provides a network-based second-order formula for the distance to the efficient frontier, generalizing Hsieh–Klenow style misallocation formulas.
- **Conceptual result.** Changes in allocative efficiency over time and the level of allocative efficiency relative to the frontier are different objects. They can move in opposite directions.
- **Applied result.** In the U.S. data, reallocation toward high-markup firms appears to have made a large positive contribution to measured TFP growth, while the static cost of markup dispersion is also large and rising.

7.2 Economic intuition

- Efficient economies are special because the initial allocation is already optimal. Then first-order reallocations do not matter, and Hulten’s theorem goes through.
- Distorted economies are different because producers face wedges between prices and marginal costs. A shock can therefore change output not only by moving the production frontier, but also by shifting resources toward or away from under- or over-supplied uses.

- Production networks amplify this logic. A firm’s macro importance depends not just on its own sales, but on how its supply chain sits inside the web of wedges and substitution elasticities.
- That is why the paper’s applied message is subtle but important: reallocation toward high-markup firms can be growth-enhancing in the short accounting sense while monopoly distortions remain socially costly in the static welfare sense.

7.3 Takeaway

This paper is a bridge between growth accounting and the modern misallocation literature. Its main insight is that once markets are distorted, aggregation is no longer just about sales shares; it is also about how shocks reshape the allocation of resources across distorted supply chains. The U.S. markup application is ultimately only a proof of concept, but it shows that these theoretical distinctions are not small technicalities. They change how we measure TFP growth, how we think about reallocation, and how large we think the welfare cost of monopoly distortions really is.